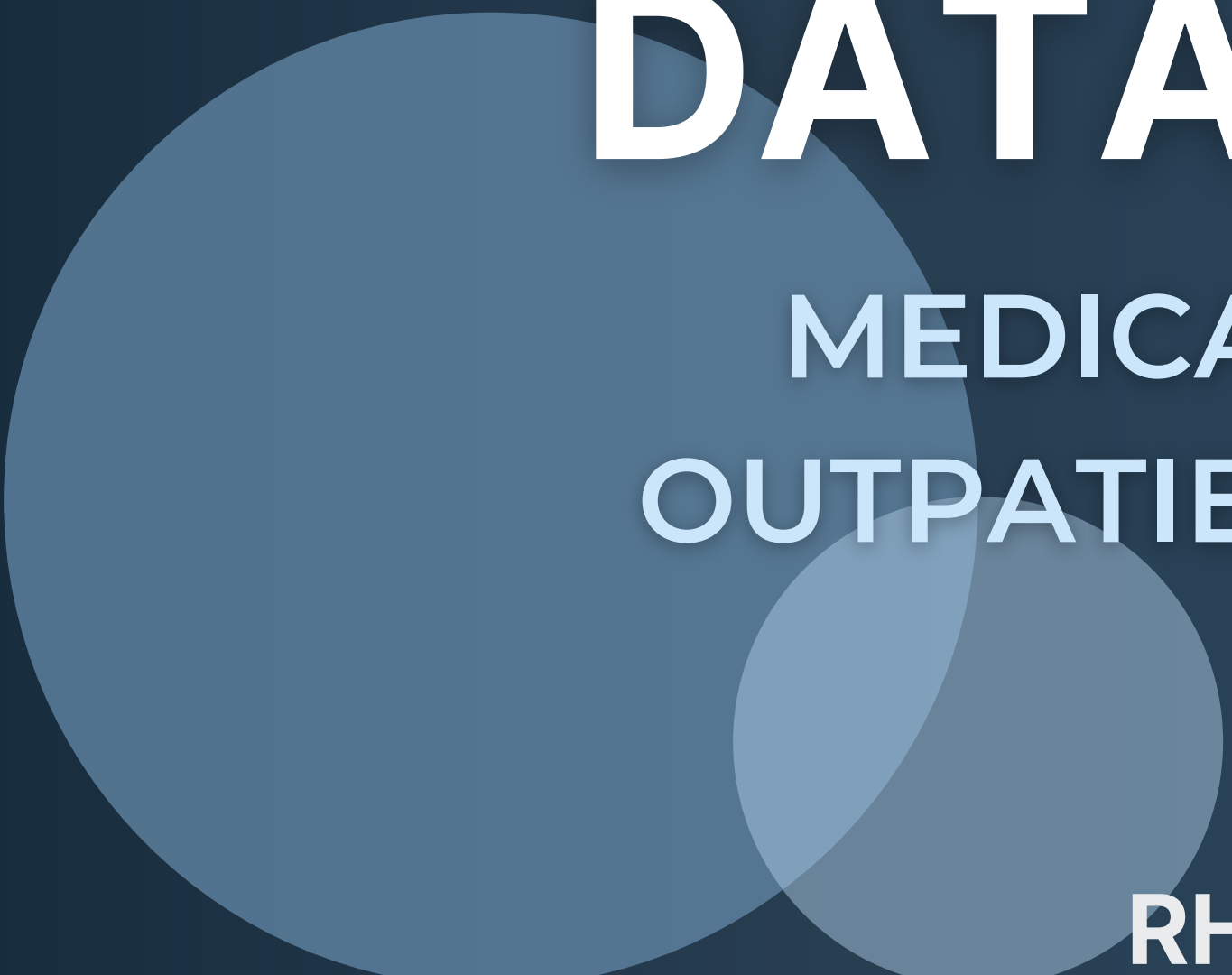


The background features three overlapping circles in shades of blue. One large circle is on the left, and two smaller circles overlap it towards the center.

DATA ANALYSIS

MEDICARE BENEFICIARY &
OUTPATIENT CLAIMS ANALYSIS
(2009)

RHYTHM CHAUDHARY



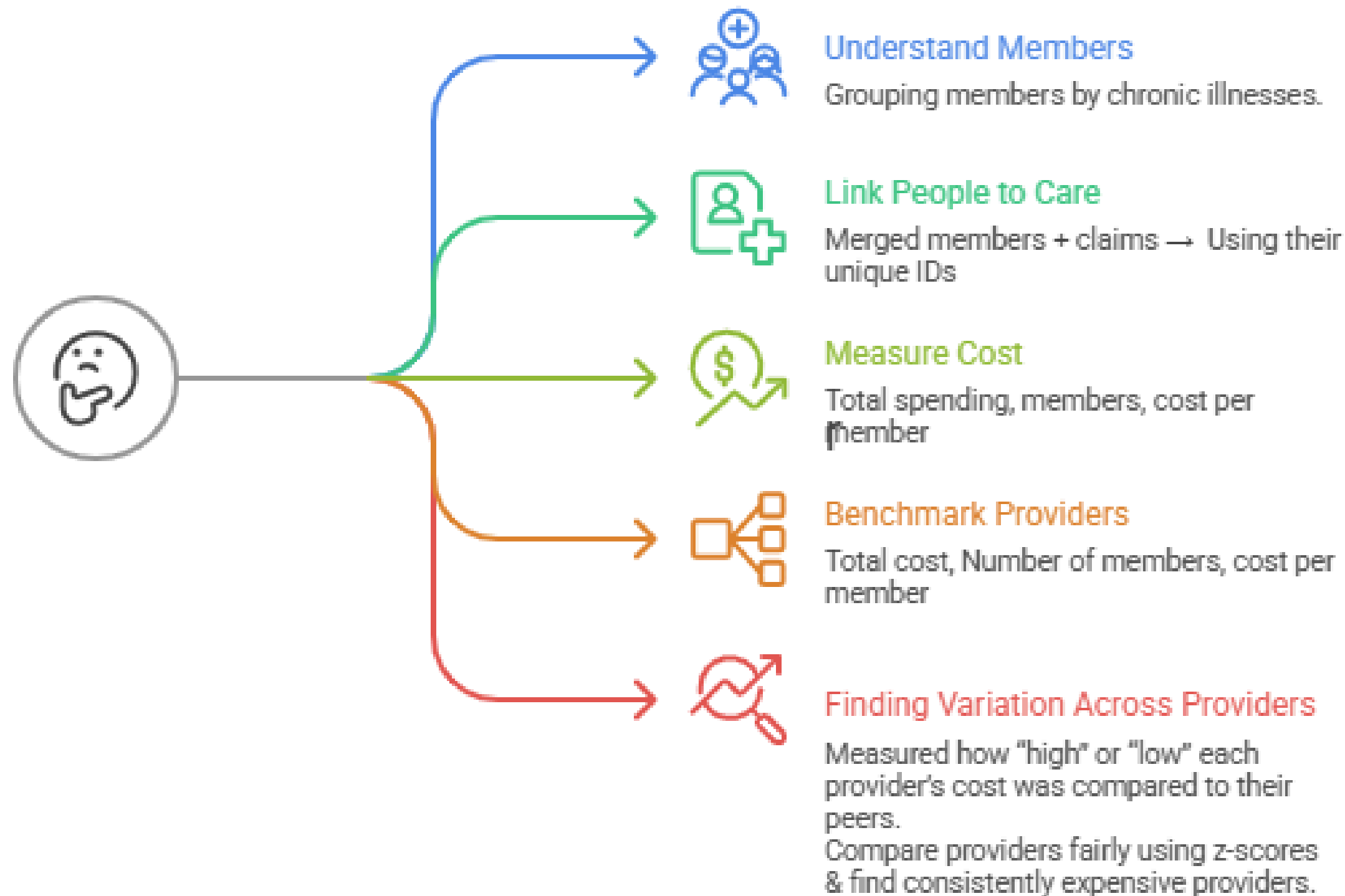
DATA USED:

- 2009 Beneficiary Summary File (chronic condition indicators)
- 2008–2010 Outpatient Claims (filtered to 2009)

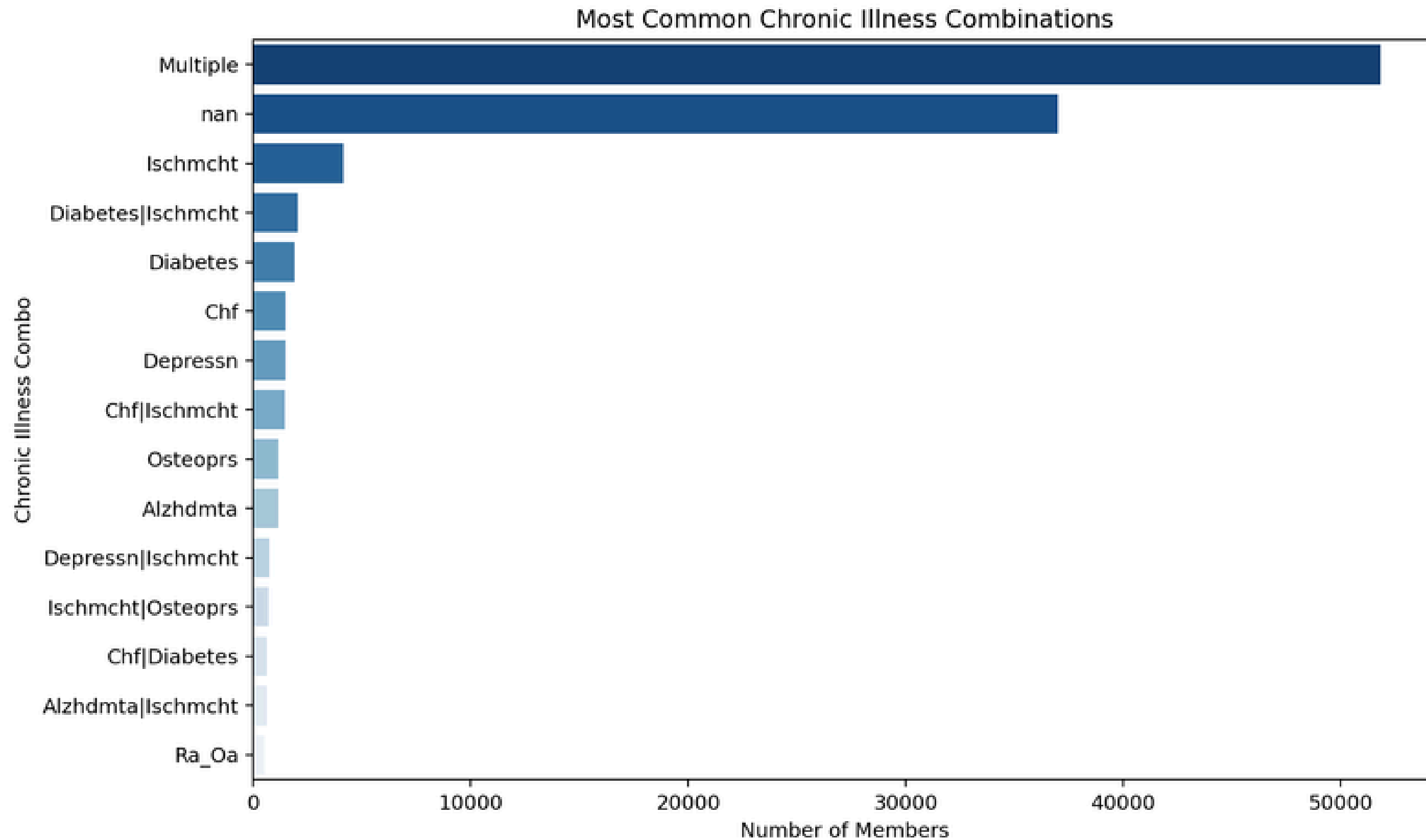
TOOLS USED:

Python (Pandas, Matplotlib, Seaborn)

APPROACH: We connected member's chronic conditions to their real care costs and benchmarked providers to reveal where spending is highest and where practice patterns vary.

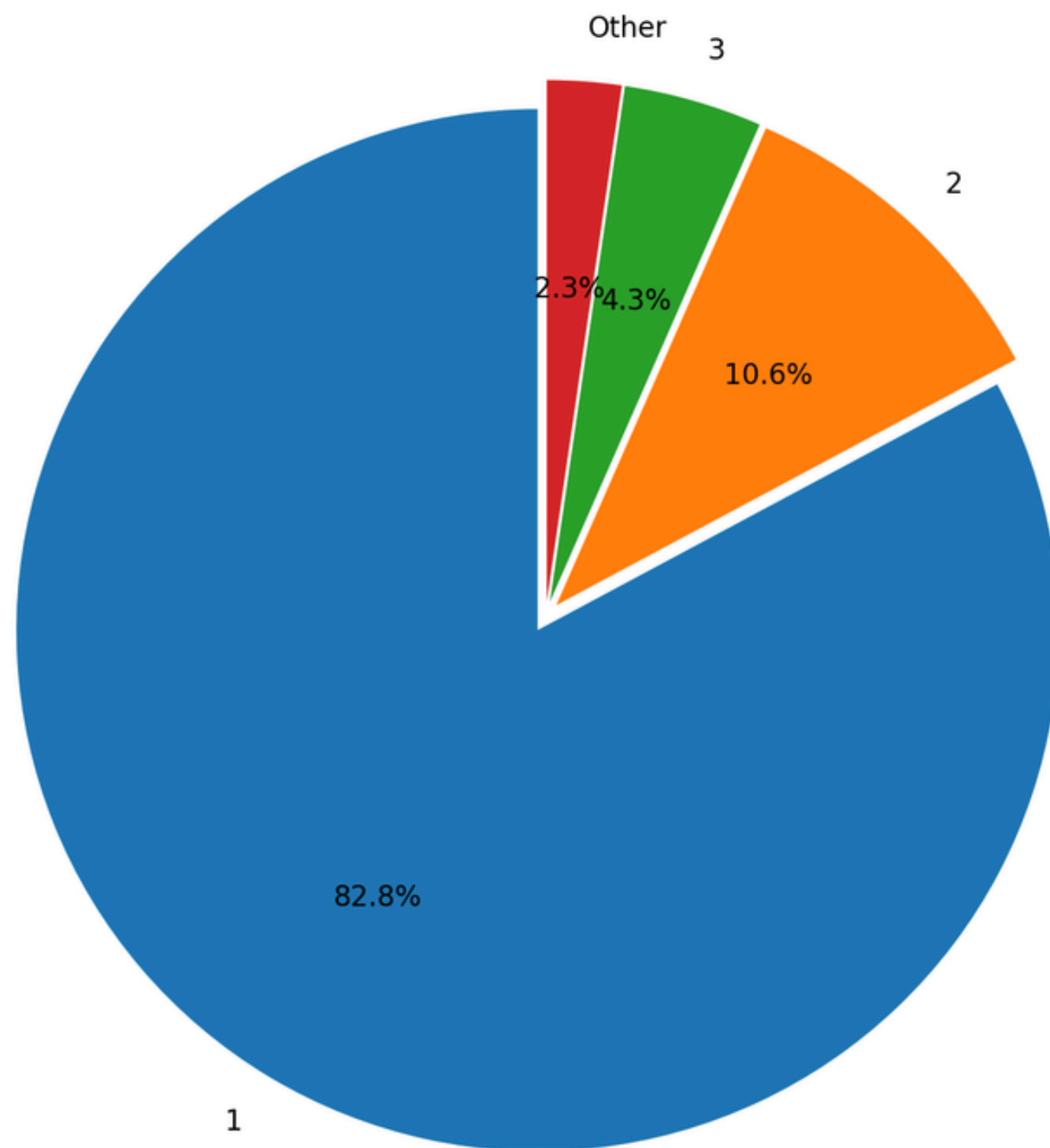


#1: Most Common Chronic Illness Combination



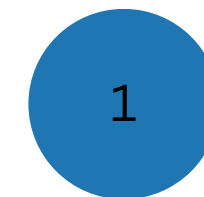
Most members fall into Multiple chronic condition categories. [Multiple =>3]
Thus, represent the largest clinical burden and require coordinated care.

Race Distribution

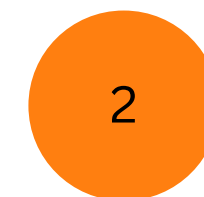


#2: Race Distribution

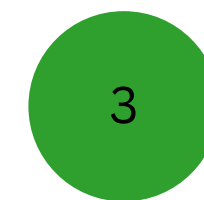
White beneficiaries denoted by "1" represent the largest groups.



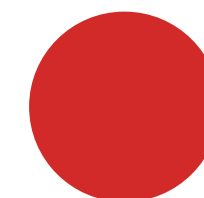
White



Black/African American

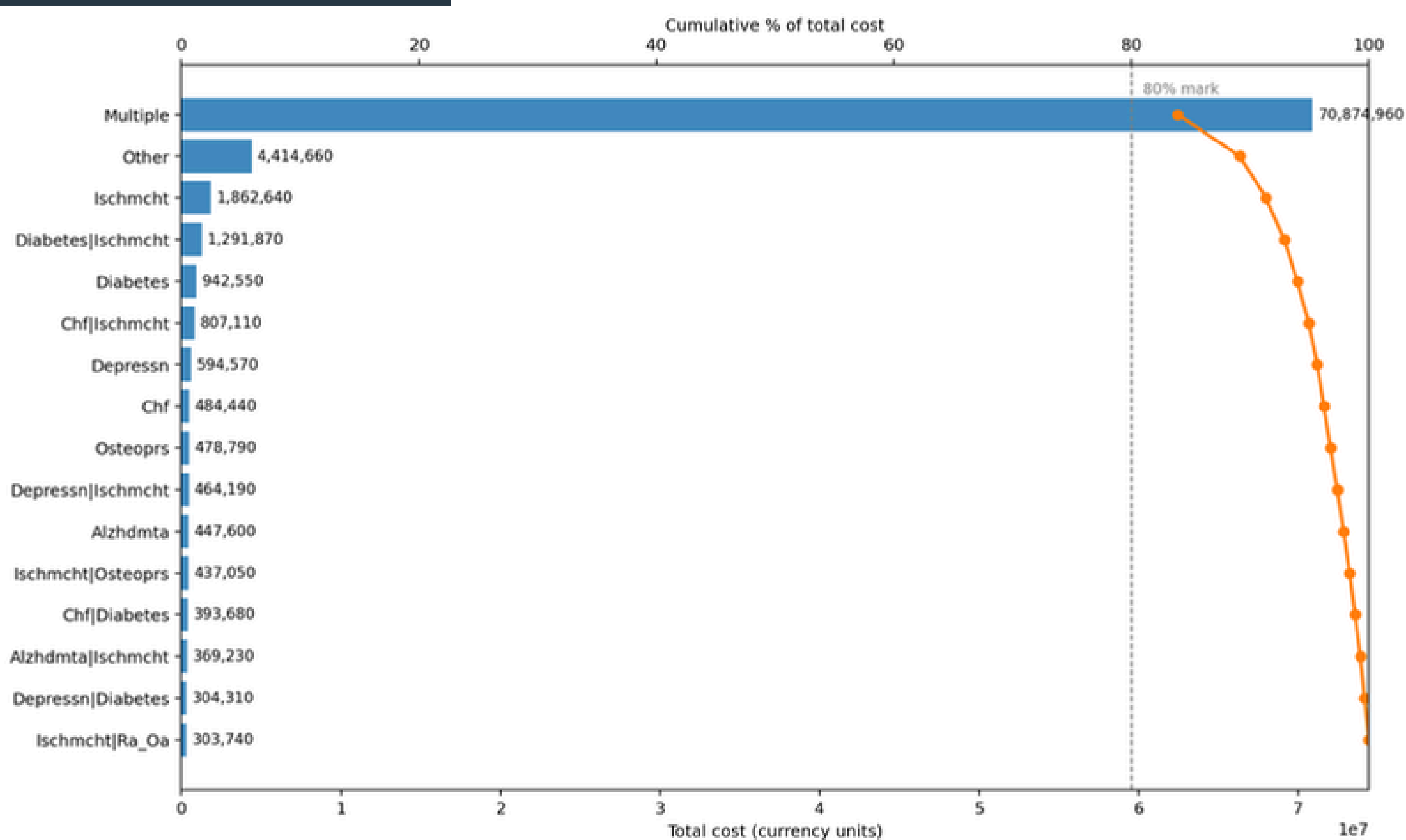


Asian



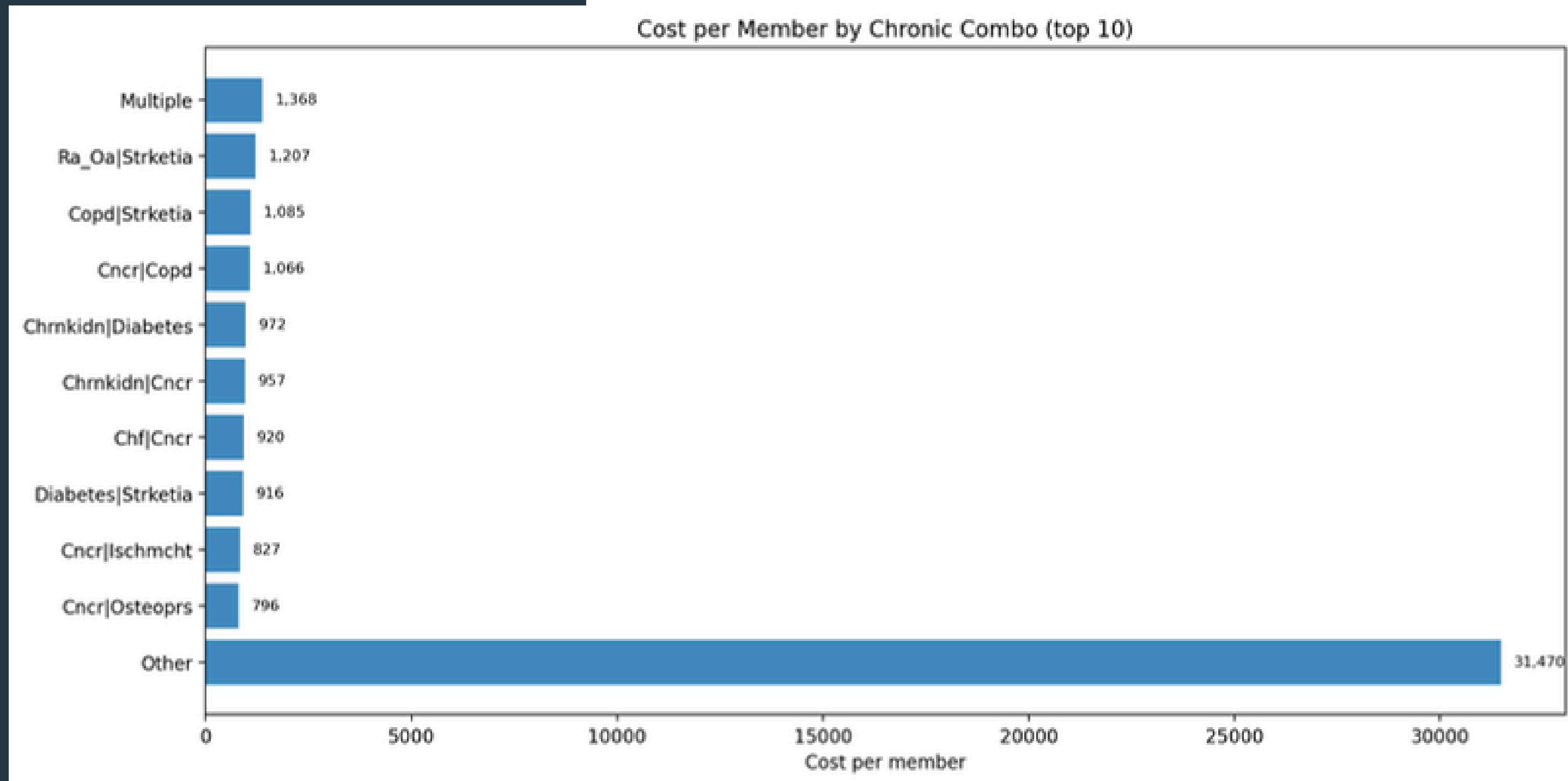
Others

#3: Which Chronic Groups Drive the Highest Cost?



‘Multiple’ drive the highest total outpatient costs.

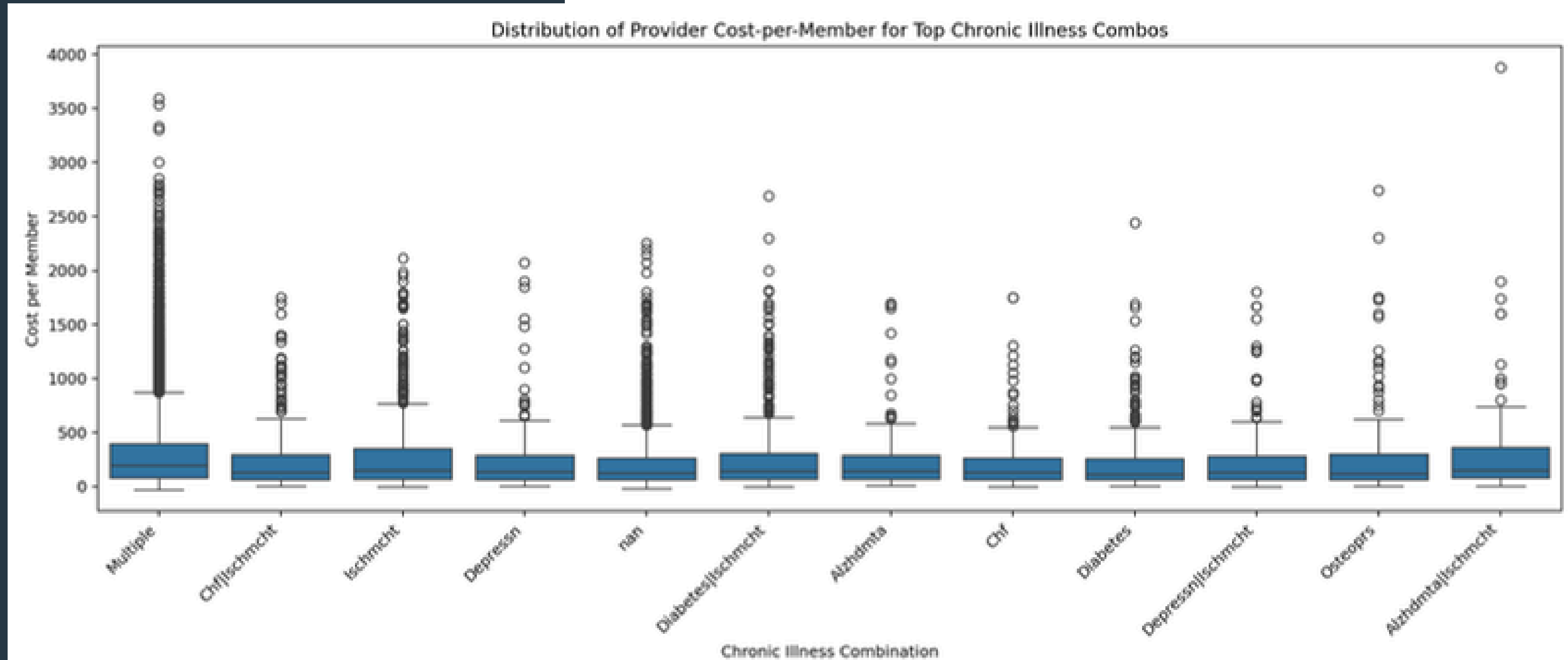
#4: Cost per Member by Chronic Group



Conditions such as Srketia, COPD, Cancer and Diabetes combinations show significantly higher per-member costs due to complexity and utilization.

NOTE: 'Other' includes all the chronic groups having "cost per member" < 796

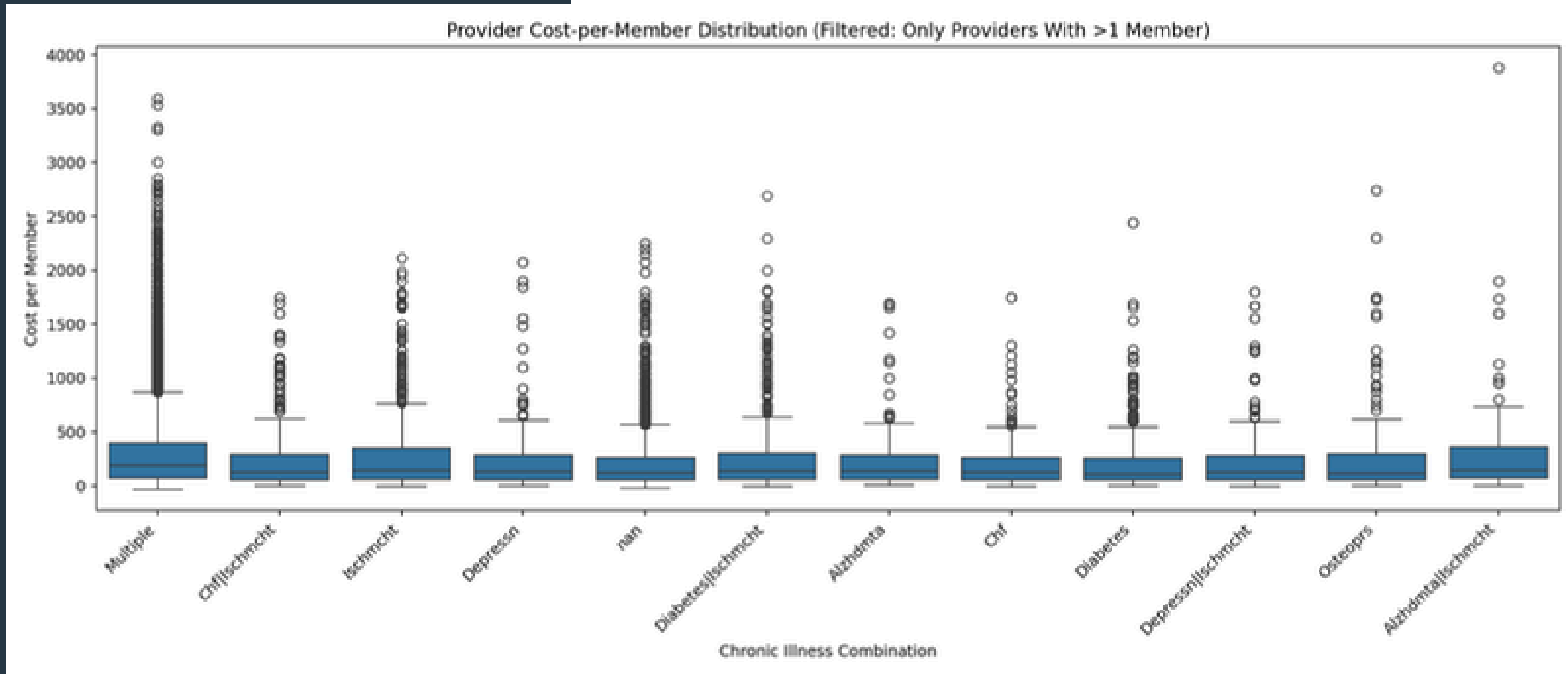
#5: Cost-per-Provider Distribution Across Chronic Illness Combinations



All condition groups have some high-cost members, but the “Multiple” group has the most extreme cases.

Certain combinations show more high-cost cases, indicating higher care complexity.

Provider Cost-per-Member Distribution After Removing 1-Member Provider-Condition Pairs



After removing provider-condition pairs with only 1 member, the cost-per-member distributions become far more stable and meaningful, revealing true provider variation instead of noise driven by tiny sample sizes.

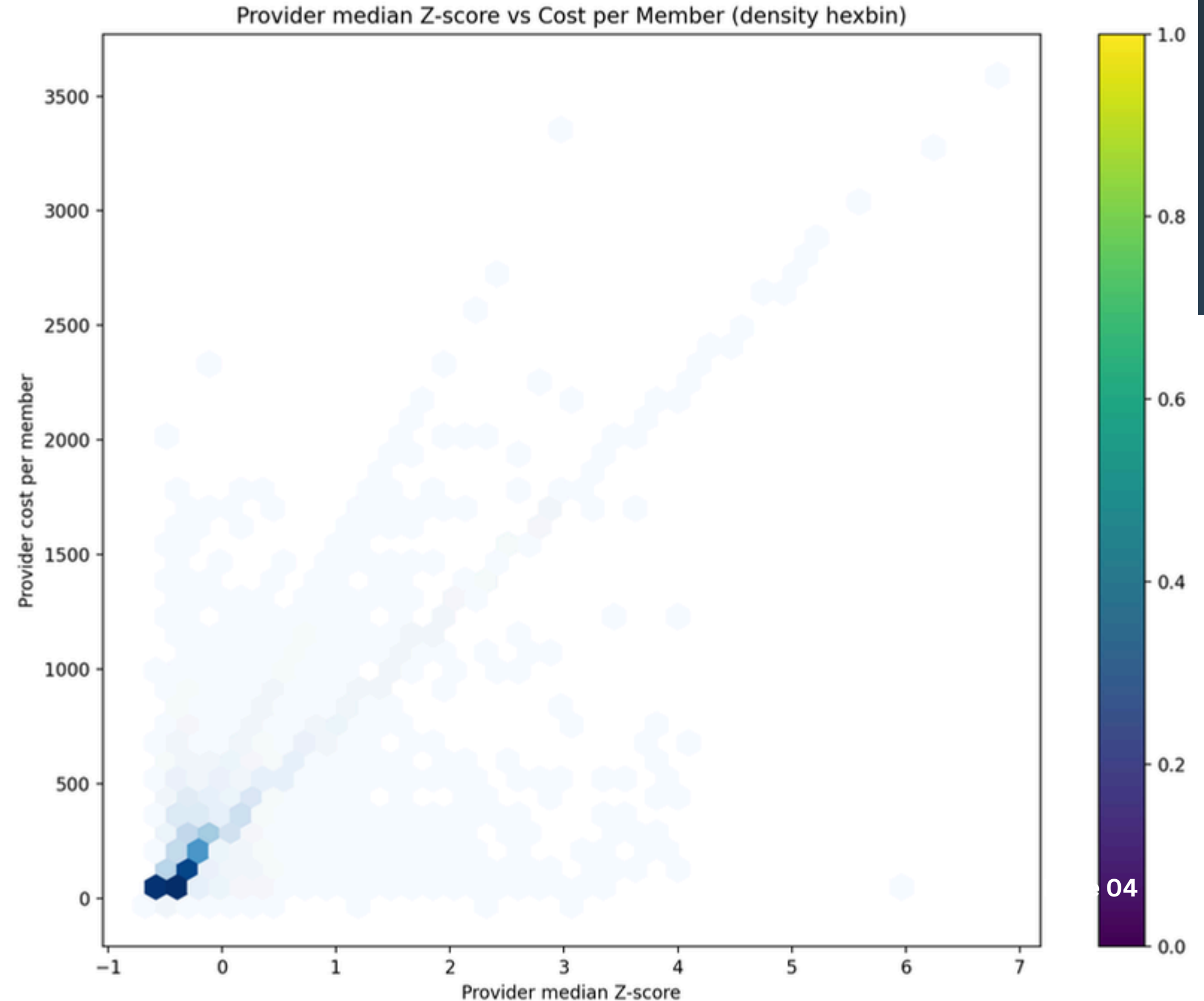
#6: Provider Outlier Detection

Approach:
BENCHMARKING PROVIDERS

For each chronic group:

- Standardized cost per member using z-scores
- Positive z-score → more expensive than peers
- Negative z-score → more efficient

Then took median z-score per provider across all groups.



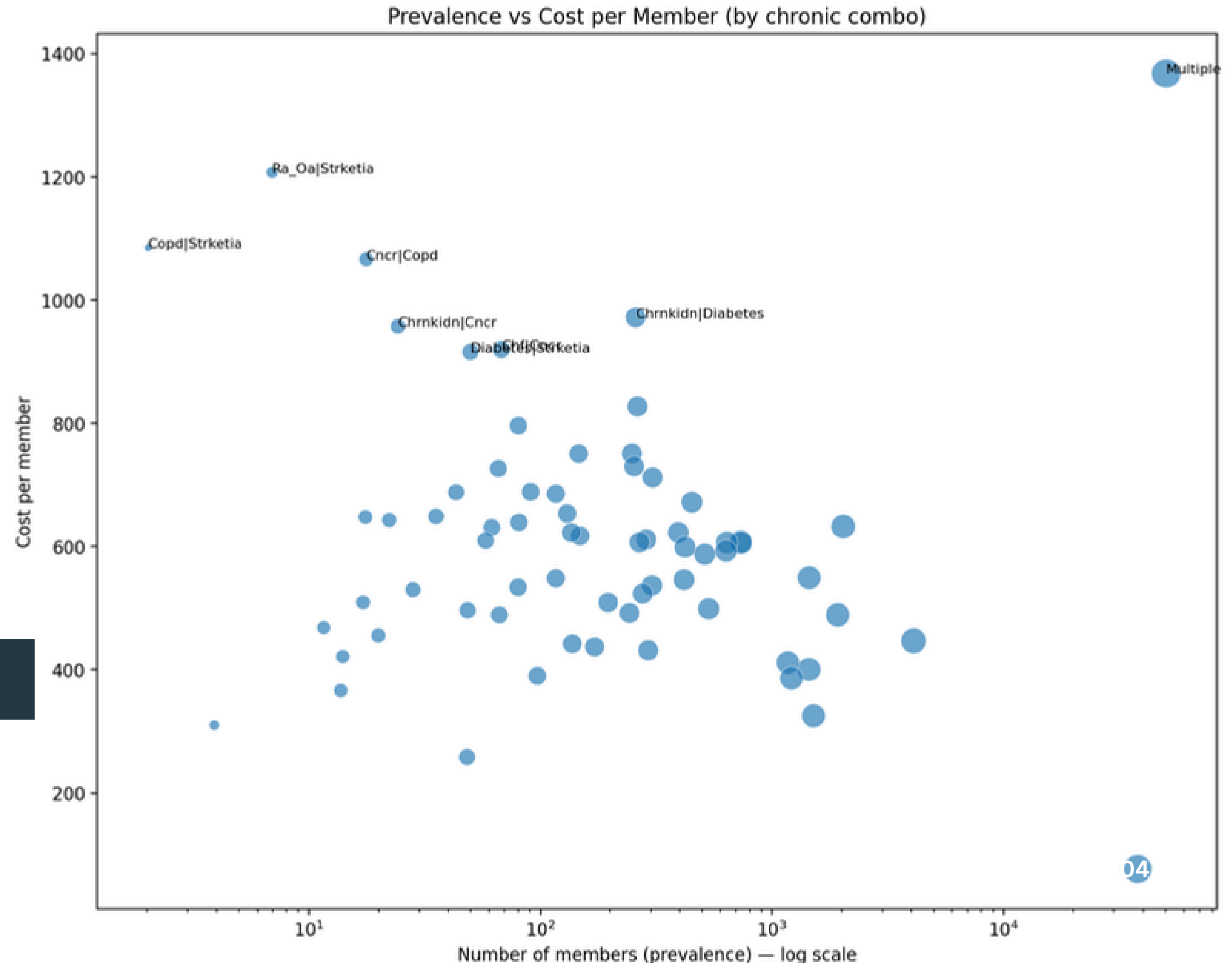
#7: Provider Scatter Plot

Right = More Members,
Up = Higher Cost Per Person
Bubble size = Total Cost Impact

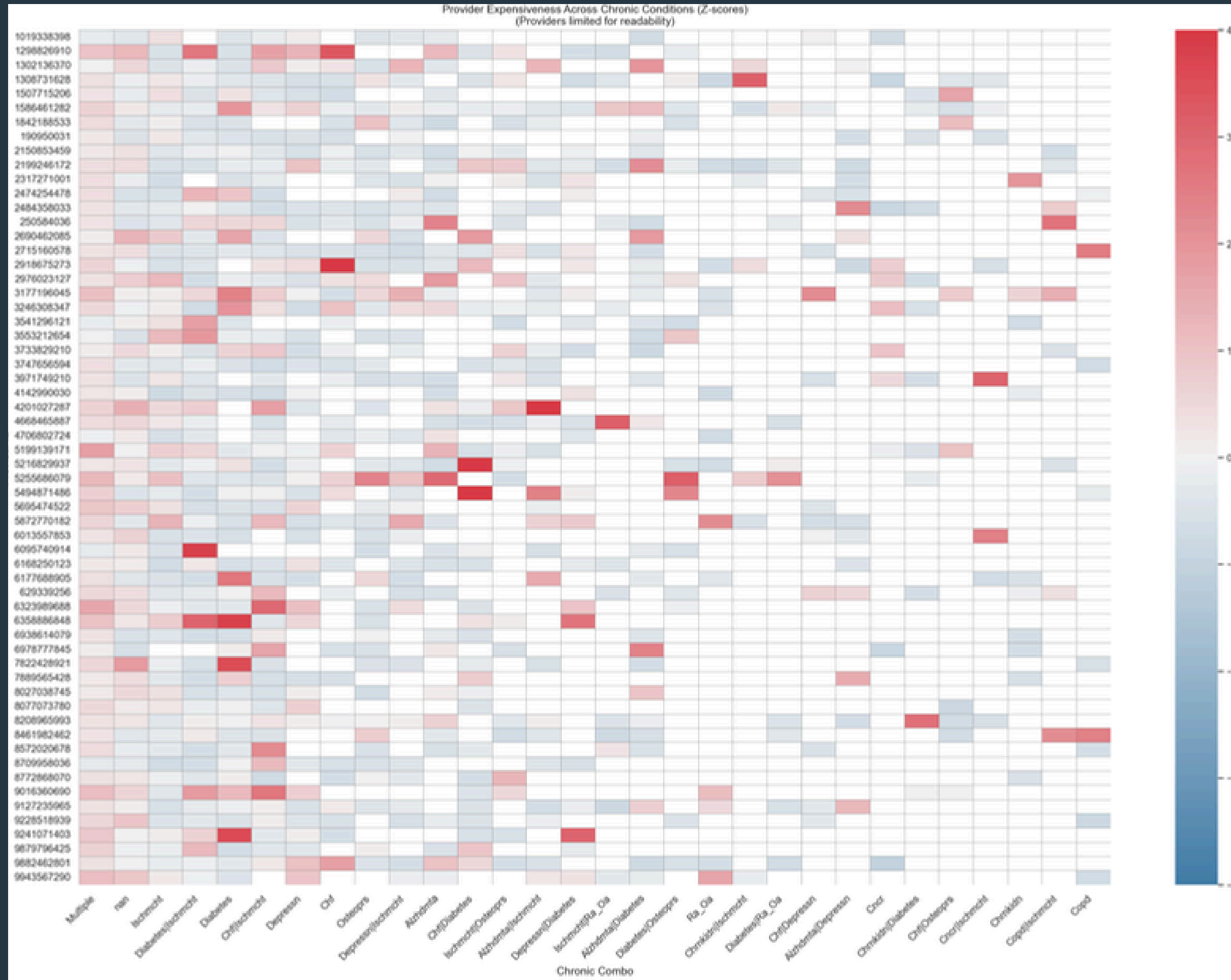
Key Observations:

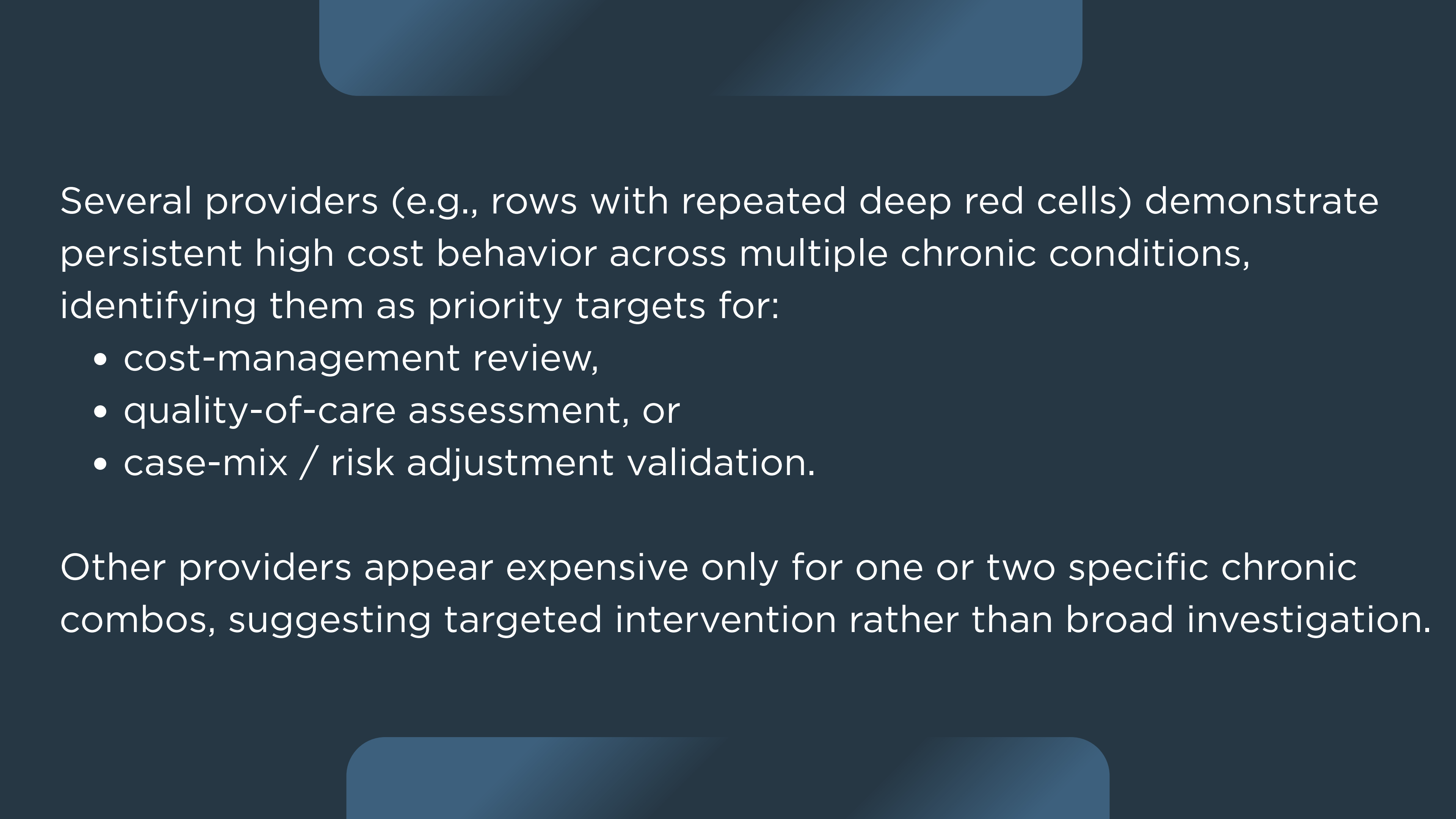
Rare condition groups often have very high costs per member.

The “Multiple” group is both common and costly, making it a major cost driver.



Identifying Providers Consistently Expensive Across Chronic Conditions





Several providers (e.g., rows with repeated deep red cells) demonstrate persistent high cost behavior across multiple chronic conditions, identifying them as priority targets for:

- cost-management review,
- quality-of-care assessment, or
- case-mix / risk adjustment validation.

Other providers appear expensive only for one or two specific chronic combos, suggesting targeted intervention rather than broad investigation.

MY RECOMMENDATIONS

Use predictive modeling to identify members at risk of transitioning from single to multiple chronic conditions.

Study links like OptumRx can cut avoidable medical spend by linking medication adherence to outpatient cost patterns, as a PBM.

Predict which outpatient encounters could be diverted to lower-cost sites of care (urgent care, telemedicine).